Alpha Equivalence for Named and Nameless Lambda Terms

A formalization in Lean

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Same thing, different names



$$\text{Morning star} \quad \mapsto \quad \text{Venus} \quad \leftarrow\!\shortmid \quad \text{Evening star}$$

$$\lambda a.\, a\, b \quad \mapsto \quad [\lambda a.\, a\, b]_{\alpha} \quad \leftarrow\!\shortmid \quad \lambda c.\, c\, b$$

Frege, *Über Sinn und Bedeutung* (1892)

Why binders?

- Standard canon:
  - Church, *The Calculi of Lambda-Conversion* (1941)
  - Curry and Feys, *Combinatory Logic* (1958)
  - de Bruijn, *Lambda calculus notation with nameless dummies* (1972)
  - Barendregt, *The Lambda Calculus* (1984)
  - Pierce, *Types and Programming Languages* (2002)
  - the POPLmark challenge (2005)
  - ...
- Proof assistants have to pick a representation for terms:
  - de Bruijn indices: Rocq, Agda
  - locally nameless: Isabelle, Lean
  - named: HOL Light
  - nominal techniques: Nominal Isabelle
- Compilers: global names and linking.

Binders in practice: CompCert linking

```
From: Xavier Leroy <xavier.leroy@college-de-france.fr>
To: compcert-users@inria.fr
Subject: Re: [compcert-users] Pass-introduced runtime helpers and
         separate compilation (TransfLink)
Date:  Wed, 21 Oct 2026 18:54:20 +0200
```

```
> So a shared name must be public in both units. The code-comment above
> it says that same-name statics would need a renaming of globals that
> CompCert does not have.

Indeed.  That's perhaps the greatest limitation of CompCert's model of
linking.  However, renaming static globals requires a whole lot of code
(to perform the renaming uniformly in abstract syntax trees) and of
proofs (to show invariance of operational semantics w.r.t. renamings).
So much code and so much proof that a completely different approach,
based on nominal logics for example, might be warranted...
```

The two papers

Crole (2012), *Alpha equivalence equalities*

$$\sim_p = \sim_{p\#} = \sim_p^1 = \sim_r = \sim_{r\#}$$

Five definitions of α from the literature are the same relation.

Norrish and Vestergaard (2007), *de Bruijn terms really do work*

$$\langle \Lambda_\alpha, \to_\beta \rangle \;\cong\; \langle \mathrm{dB}, \to_\mathrm{d} \rangle$$

Named terms up to α and de Bruijn terms are isomorphic reduction systems.

Λ_α may quotient by any of the 5; CSLib uses $\sim_p$.

Plan

Section in the Lean file	Topic
`S1_NamedTerms`	terms, rename, swap, subst
`S2_Crole`	5 relations, 1 theorem
`S3_Isomorphism`	$\Lambda_\alpha \cong \mathrm{dB}$ in use, Church–Rosser
if time permits: how it was proved	
`S4_Route`	via dB'
`S5_Permutations`	lifting is a permutation
`S5a_Nominal`	what nominal sets were good for
`B1–B3`	backup: departures from the papers

Permutation-based α (Crole 2012, Defs. 3.1–3.3)

$$\frac{}{a \sim a}\ \text{ATOM} \qquad \frac{M_1 \sim M_1' \quad M_2 \sim M_2'}{M_1\, M_2 \sim M_1'\, M_2'}\ \text{PCG}$$

$$\frac{(z\ a) \cdot M \sim (z\ b) \cdot M'}{\lambda a.\, M \sim \lambda b.\, M'}\ \text{PI}$$

Relation	rule	side condition	Lean
$\sim_p$	PI	$z \notin \{a, b\} \cup \mathrm{vars}(M) \cup \mathrm{vars}(M')$	`AlphaEquiv`, $=\alpha$
$\sim_{p\#}$	$\mathrm{PI}_\#$	$z \notin \{a, b\} \cup \mathrm{fv}(M) \cup \mathrm{fv}(M')$	`AlphaEquivPFresh`
$\sim_p^1$	PI_1	$z \notin \mathrm{vars}(M) \cup \mathrm{vars}(M')$	`AlphaEquivP1`

vars: all atoms, binders included. fv: free atoms. $(z\ a) \cdot M$: swap z and a everywhere.

Renaming-based α (Crole 2012, Defs. 3.4, 3.5)

$$\frac{}{M \sim M}\ \text{REF} \qquad \frac{M_1 \sim M_2}{M_2 \sim M_1}\ \text{SYM} \qquad \frac{M_1 \sim M_2 \quad M_2 \sim M_3}{M_1 \sim M_3}\ \text{TRS}$$

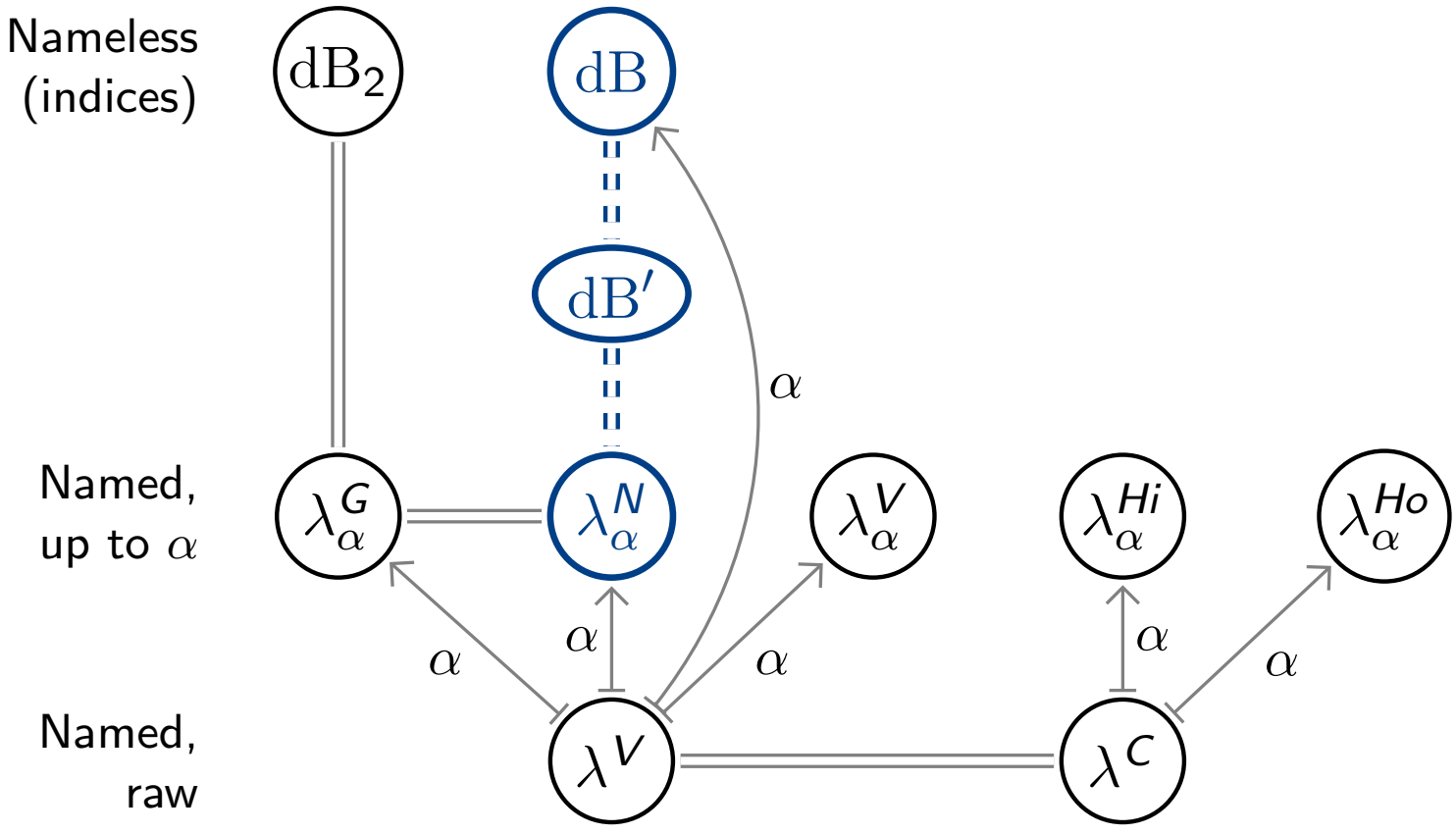
$$\frac{M_1 \sim M_1' \quad M_2 \sim M_2'}{M_1\, M_2 \sim M_1'\, M_2'}\ \text{PCG} \qquad \frac{M \sim M'}{\lambda a.\, M \sim \lambda a.\, M'}\ \text{BCG}$$

$$\frac{}{\lambda a.\, M \sim \lambda a'.\, M[a := a']}\ \alpha$$

Relation	rule	side condition	Lean
$\sim_r$	α	$a' \notin \{a\} \cup \mathrm{vars}(M)$	`AlphaEquivR`
$\sim_{r\#}$	$\alpha_\#$	$a' \notin \{a\} \cup \mathrm{fv}(M)$	`AlphaEquivRFresh`

$M[a := a']$: capture-avoiding substitution.

The network (Norrish and Vestergaard 2007, Fig. 1)



Double lines: isomorphisms. Arrows: α-quotients. Dashed: the paper's results.
Blue: in CSLib.

β-reduction on named terms up to α

$\Lambda_\alpha = \Lambda/{=_\alpha}$, with the operations lifted from representatives:

$$[M]_\alpha\,[N]_\alpha = [M\,N]_\alpha \qquad \lambda a.\,[M]_\alpha = [\lambda a.\,M]_\alpha$$

$$[M]_\alpha[a := [N]_\alpha] = [M[a := N]]_\alpha$$

$$\frac{}{(\lambda a.\,M)\,N \to_\beta M[a := N]} \qquad \frac{M \to_\beta N}{\lambda a.\,M \to_\beta \lambda a.\,N}$$

$$\frac{M \to_\beta N}{M\,Z \to_\beta N\,Z} \qquad \frac{M \to_\beta N}{Z\,M \to_\beta Z\,N}$$

Lean: `BetaAlpha`, notation $\to_{\beta\alpha}$.

β-reduction on de Bruijn terms (Nipkow)

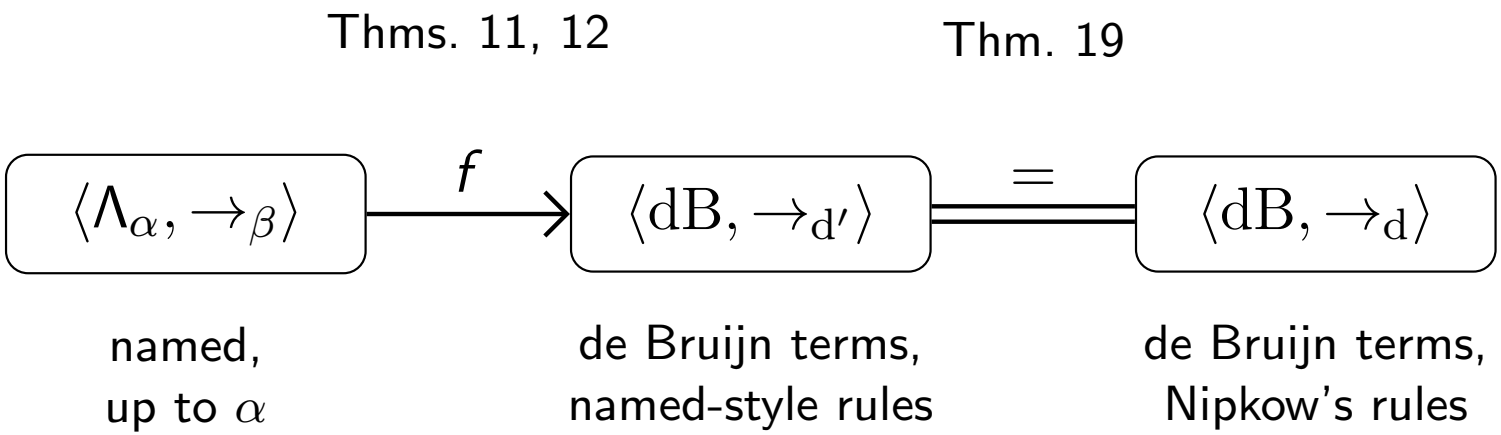
De Bruijn terms $t ::= i \mid t\ t \mid \lambda.\ t$

$$\frac{}{(\lambda.\ t)\ u \to_{\mathrm{d}} \mathsf{nsub}\ u\ 0\ t} \qquad \frac{t \to_{\mathrm{d}} u}{\lambda.\ t \to_{\mathrm{d}} \lambda.\ u} \qquad \frac{t \to_{\mathrm{d}} u}{t\ s \to_{\mathrm{d}} u\ s} \qquad \frac{t \to_{\mathrm{d}} u}{s\ t \to_{\mathrm{d}} s\ u}$$

$\mathsf{nsub}\ u\ k\ t$: replace index k by u, decrement the free indices above k.

Lean: `Beta`, notation $\to_\beta$.

The route (Norrish and Vestergaard 2007)



Theorem 20: $M \to_\beta N \iff f(M) \to_{\mathrm{d}} f(N)$